

ETH ZÜRICH 5-YEAR PREPARATION GUIDE AND QUESTION BANK

Comprehensive Study Plan, Tactics, and a 160-Question Bank (Part 1)

Author: FF_Weath

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Special Edition: For Engineering and Computer Science Applicants

This Document is Part of a Series

This file is the **1st part** of the **ETH Zürich 5-Year Preparation Guide and Question Bank** series. The complete series:

- **Part 1:** General tactics, study plan, 5-year roadmap, mathematics and foundational physics.
- **Part 2:** Physics (electricity, magnetism, thermodynamics, waves), biology, chemistry, German language.
- **Part 3:** Advanced mathematics, advanced physics, academic German reading and writing.
- **Part 4:** Mixed mock exams and exam strategies.

Important: It is strongly recommended to download and study the entire series, not just this part.

Recommendation: Read the general tactics and study plan in this part first. Then use the other parts to focus on the relevant subjects.

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Section 1: Problem-Solving Tactics

The ETH Zürich (and Swiss Matura) exams test not formula memorization, but the ability to model systems and think like an engineer.

1. Divide and Conquer

Break a complex problem into independent subsystems.

Physics example: For a cylinder rolling down an inclined plane, write translational kinematics (Newton) and rotational dynamics (torque) as completely separate equations, then link them with $v = \omega r$.

2. Reverse Engineering

Work backward from the desired final result.

Math example: If asked for the area between two curves, reason: "I need a definite integral $\rightarrow$ I need limits $\rightarrow$ I need intersection points from $f(x) = g(x)$."

3. Debug Your Own Work

Apply the "edge-case testing" mindset from software.

Math example: When you derive a general formula, plug in $x = 0$, $x = 1$, or $x \rightarrow \infty$ to test whether the result makes sense.

4. Visualization

Geometry and mechanics problems cannot be solved without drawings. Free-body diagrams (FBD) and function graphs dramatically reduce error rates.

5. Dimensional Analysis

Check the consistency of units in your final formula. *Physics example:* If you derived an energy equation, the unit must be $kg \cdot m^2 / s^2$ (Joule).

Section 2: Study Strategies

1. Deep Work

Uninterrupted focus is essential. Use the "Pomodoro" technique (50 min focus, 10 min break) and completely switch off your phone.

2. Active Recall & Spaced Repetition

Reading formulas is passive. Use **Anki** or **Notion** to periodically test yourself.

3. Feynman Technique

The best way to understand a topic is to explain it to someone who knows nothing about it.

4. Past Paper Analysis

You must solve ETH's previous "Aufnahmeprüfung" exams. Every wrong answer should be recorded in a dedicated "Error Log".

5. Avoiding Burnout

Mental health is critical in a 5-year process. Take one full day off (Sunday) each week.

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Section 3: Oral Exam Strategies

In the Swiss system, the "Mündliche Prüfung" (oral exam) is decisive. You must narrate your thought process.

1. Narrating the Thought Process

Even if you take a wrong turn, if the examiner hears your reasoning, you get points.

- **EN:** "First, I will resolve the forces into their components, and then..."
- **DE:** "Zuerst werde ich die Kräfte in ihre Komponenten zerlegen, und dann..."

2. Buying Time

- **EN:** "Let me quickly sketch this out to visualize the problem better."
- **DE:** "Lassen Sie mich das kurz skizzieren..."

3. Self-Correction

- **EN:** "Wait, my dimensional analysis fails here. I must have dropped a factor of x ."
- **DE:** "Moment, meine Einheitenprüfung schlägt fehl..."

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Section 4: Topics and Priority Matrix

Subject / Topic	English / German	Priority	Notes
Math: Derivatives & Integrals	Calculus / Differential- und Integralrechnung	HIGH	Optimization, volumes, Taylor series. Proofs may be required.
Math: Vectors & 3D Geometry	Vectors, 3D Geometry / Vektorgeometrie	HIGH	Line-plane intersections, projection, cross product.
Math: Complex Numbers	Complex Numbers / Komplexe Zahlen	Medium	Euler form, polar coordinates, De Moivre.
Math: Probability	Probability / Wahrscheinlichkeit	Medium	Binomial, Bayes, conditional probability.
Physics: Mechanics	Mechanics / Mechanik	HIGH	Energy conservation, angular momentum, friction.
Physics: Electromagnetism	Electromagnetism / Elektromagnetismus	HIGH	Kirchhoff, Lorentz force, induction.
Physics: Thermodynamics & Waves	Thermodynamics, Waves	Medium	Ideal gas, heat engines, Snell's law.
Languages: Academic Reading/Writing	Academic Reading & Essay	HIGH	Article analysis, argumentative essay (Erörterung).

This section is yours. Click any cell and type. Editable in the browser. When you print → save as PDF, your entries are preserved.

Day	Subject Focus	Daily Task	Time
Monday			30 min
Tuesday			30 min
Wednesday			30 min
Thursday			30 min
Friday			30 min
Saturday			30 min
Sunday			Rest

Month	Goal	Notes
September		
October		
November		
December		
January		
February		

Write down every question you got wrong, then re-solve it two weeks later.

Date	Question / Topic	Error Cause	Review
			☐
			☐
			☐
			☐
			☐

Section 6: Resources

Books

- **Math (DE):** *Mathematik für Ingenieure und Naturwissenschaftler* – Lothar Papula (Vol. 1 & 2).
- **Physics (EN):** *Fundamentals of Physics* – Halliday, Resnick, Walker.
- **Physics (DE):** *Tipler/Mosca: Physik für Wissenschaftler und Ingenieure*.

Websites & Archives

- ETH Zürich Aufnahmeprüfung archive: ethz.ch
- Mathe-Physik-Aufgaben: mathe-physik-aufgaben.de
- Paul's Online Math Notes: tutorial.math.lamar.edu
- GitHub: github.com/janikschuettler/eth-entrance-exam

Tools

- **Geogebra 3D:** Vector and 3D geometry visualization.
- **Anki:** Spaced repetition flashcards.
- **Desmos:** Function graphs and derivative/integral visualization.

Section 7: Question Bank (Part 1 – 40 Questions)

Note: All questions are **original and adapted** problems in the *ETH/Matura* style.

A. Algebra, Polynomials, and Functions

Q1 **Hard** **Algebra** **12m**

EN: Show that $P(x) = x^4 - 6x^3 + 13x^2 - 12x + 4$ is a perfect square and write it in the form $Q(x)^2$.

DE: Zeigen Sie, dass $P(x)$ ein vollständiges Quadrat ist, und schreiben Sie es in der Form $Q(x)^2$.

 **Strategy:** Assume $P(x) = (x^2 + ax + b)^2$ and match coefficients.

Solution: $P(x) = (x^2 - 3x + 2)^2 = (x - 1)^2(x - 2)^2$.

 Working space

Q2 **Easy** **Algebra** **5m**

EN: Simplify $\frac{2x}{x^2-1} - \frac{1}{x-1}$.

DE: Vereinfachen Sie $\frac{2x}{x^2-1} - \frac{1}{x-1}$.

Solution: $\frac{1}{x+1}$ ($x \neq 1$).

 Working space

Q3 **Hard** **Inequality** **10m**

EN: Solve the inequality $0 < \frac{x+2}{x-1} \leq 1$.

DE: Lösen Sie die Ungleichung $0 < \frac{x+2}{x-1} \leq 1$.

Solution: $(-\infty, -2)$.

 Working space

Q4 Medium Inverse Function 7m

EN: Find the inverse of $f(x) = \sinh(x) = \frac{e^x - e^{-x}}{2}$.

DE: Bestimmen Sie die Umkehrfunktion von $f(x) = \sinh(x)$.

Solution: $f^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$.

 Working space

Q5 Hard Logarithms 8m

EN: Solve $\log_{10}(x) + \log_{10}(x - 0.9) = -1$.

DE: Lösen Sie $\log_{10}(x) + \log_{10}(x - 0.9) = -1$.

Solution: $x = 1$ (domain $x > 0.9$).

 Working space

Q6 Easy Inverse Function 4m

EN: If $f(x) = \frac{2x+3}{x-4}$, find $f^{-1}(5)$.

DE: Wenn $f(x) = \frac{2x+3}{x-4}$, berechnen Sie $f^{-1}(5)$.

Solution: $f^{-1}(5) = \frac{23}{3}$.

 Working space

Q7 Medium Limits 6m

EN: Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin(x)}{x^3}$.

DE: Berechnen Sie $\lim_{x \rightarrow 0} \frac{x - \sin(x)}{x^3}$.

Solution: $\frac{1}{6}$ (L'Hôpital or Taylor).

 Working space

Q8 **Hard** Optimization 12m

EN: Find the dimensions of the rectangle with maximum area inscribed under the parabola $y = 4 - x^2$ above the x-axis.

DE: Bestimmen Sie die Abmessungen des Rechtecks mit maximalem Flächeninhalt unter der Parabel $y = 4 - x^2$.

Solution: $\frac{4}{\sqrt{3}} \times \frac{8}{3}$.

 Working space

B. Calculus: Limits, Derivatives, Integrals, Series

Q9 **Medium** Parametric 6m

EN: For $x = t^3 - 3t$, $y = t^2 - 2t$, find $\frac{dy}{dx}$ at $t = 2$.

DE: Für $x = t^3 - 3t$, $y = t^2 - 2t$: Berechnen Sie $\frac{dy}{dx}$ bei $t = 2$.

Solution: $\frac{2}{9}$.

 Working space

Q10 **Medium** Related Rates 7m

EN: A 10 m ladder slides with its base moving at 0.5 m/s. How fast is the top sliding down when the base is 6 m from the wall?

DE: Eine 10 m Leiter: Die Basis gleitet mit 0.5 m/s. Wie schnell gleitet die Spitze, wenn die Basis 6 m entfernt ist?

Solution: 0.375 m/s downward.

 Working space

Q11 Hard Volume 9m

EN: Find the volume generated by rotating $y = \sqrt{4 - x^2}$ on $[-2, 2]$ about the x-axis.DE: Bestimmen Sie das Volumen des Rotationskörpers von $y = \sqrt{4 - x^2}$ auf $[-2, 2]$ um die x-Achse.Solution: $\frac{32\pi}{3}$.

Working space

Q12 Medium Improper Integral 7m

EN: Evaluate $\int_0^\infty x e^{-x} dx$.DE: Berechnen Sie $\int_0^\infty x e^{-x} dx$.

Solution: 1.

Working space

Q13 Hard Maclaurin 9m

EN: Write the first 3 non-zero terms of the Maclaurin series for $\cos(x)$.DE: Schreiben Sie die ersten 3 nicht-null Terme der Maclaurin-Reihe für $\cos(x)$.Solution: $1 - \frac{x^2}{2} + \frac{x^4}{24}$.

Working space

Q14 Medium ODE 7m

EN: Solve the differential equation $\frac{dy}{dx} = 2xy$.DE: Lösen Sie die Differentialgleichung $\frac{dy}{dx} = 2xy$.Solution: $y = Ke^{x^2}$.

Working space

Q15 Medium Differentiability 6m

EN: Find a and b so that $f(x) = \begin{cases} ax^2 + 2 & x < 2 \\ x^3 - bx & x \geq 2 \end{cases}$ is differentiable at $x = 2$.

DE: Bestimmen Sie a, b , so dass f an $x = 2$ differenzierbar ist.

Solution: $a = 4.5, b = -6$.

 Working space

Q16 Hard Bayes 8m

EN: A disease affects 1% of a population. A test has 99% sensitivity and 5% false-positive rate. If a person tests positive, what is the probability they actually have the disease?

DE: Eine Krankheit betrifft 1%. Der Test hat 99% Sensitivität und 5% Falsch-Positiv-Rate. Wie groß ist die Wahrscheinlichkeit bei positivem Test?

Solution: ≈ 0.1667 (16.67%).

 Working space

Q17 Hard Wedge Dynamics 12m

EN: A frictionless wedge of mass M and angle θ rests on a horizontal floor. A block of mass m is released on it. Find the horizontal acceleration a_M of the wedge if it is free to slide.

DE: Ein reibungsfreier Keil M , Winkel θ , Block m . Bestimmen Sie die horizontale Beschleunigung a_M .

Solution: $a_M = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}$.

 Working space

Q18 Medium Series 6m

EN: Find the sum of $\sum_{n=1}^{\infty} \frac{n}{2^n}$.DE: Bestimmen Sie die Summe von $\sum_{n=1}^{\infty} \frac{n}{2^n}$.**Solution:** 2. Working space

C. Vectors, 3D Geometry, Linear Algebra

Q19 Easy Cross Product 4m

EN: Find $\vec{u} \times \vec{v}$ for $\vec{u} = (1, 2, -1)$, $\vec{v} = (3, 0, 2)$.DE: Kreuzprodukt für $\vec{u} = (1, 2, -1)$, $\vec{v} = (3, 0, 2)$.**Solution:** $(4, -5, -6)$. Working space

Q20 Medium Determinant 6m

EN: Compute $\det \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \\ 1 & -1 & 0 \end{pmatrix}$.

DE: Berechnen Sie die Determinante.

Solution: 9. Working space

Q21 **Hard** Eigenvalues 8mEN: Find the eigenvalues of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

DE: Bestimmen Sie die Eigenwerte.

Solution: $\lambda \in \{1, 3\}$.

Working space

Q22 **Medium** Line-Plane 7mEN: Find the intersection of the plane $x + y + z = 6$ with the line $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-1}$.

DE: Schnittpunkt von Ebene und Gerade.

Solution: $(7, 3, -4)$.

Working space

Q23 **Easy** Subspace 4mEN: Which of the following is NOT a subspace of $\mathbb{R}^3$?DE: Welches ist kein Unterraum von $\mathbb{R}^3$?A) $x + y + z = 0$ B) $2x - 3y = 0$ C) $x + y + z = 1$ D) $x = 0, y = 0$ **Solution:** C) $x + y + z = 1$.

Working space

Q24 **Medium** Rotation 6mEN: Write the matrix of the linear transformation in $\mathbb{R}^2$ that rotates vectors counterclockwise by 90° .DE: Matrix der 90° -Drehung gegen den Uhrzeigersinn in $\mathbb{R}^2$.**Solution:** $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Working space

Q25 **Hard** Skew Lines 10m

EN: Find the shortest distance between the skew lines $L_1 : (1, 0, 1) + t(1, 2, 1)$ and $L_2 : (2, 1, 0) + s(0, 1, 1)$.

DE: Kürzester Abstand zwischen windschiefen Geraden.

Solution: $\frac{\sqrt{3}}{3}$.

 Working space

Q26 **Medium** Rank 7m

EN: Find the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$.

DE: Bestimmen Sie den Rang der Matrix.

Solution: 2.

 Working space

D. Probability and Statistics

Q27 **Easy** Permutation 4m

EN: How many distinct 10-letter words can be formed from the letters of "MATEMATİK"?

DE: Wie viele verschiedene 10-Buchstaben-Wörter aus "MATEMATİK"?

A) $10!$ B) $\frac{10!}{2!2!2!}$ C) $\frac{10!}{3!2!}$ D) $\frac{10!}{3!}$

Solution: B) $\frac{10!}{2!2!2!}$.

 Working space

Q28 Medium Conditional 5m

EN: If $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cup B) = 0.7$, find $P(A|B)$.DE: Wenn $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cup B) = 0.7$, berechnen Sie $P(A|B)$.**Solution:** 0.4. Working space

Q29 Hard PDF 8m

EN: Given the PDF $f(x) = cx^2$ for $0 \leq x \leq 3$, find c and $E[X]$.DE: Für $f(x) = cx^2$, $0 \leq x \leq 3$: Bestimmen Sie c und $E[X]$.**Solution:** $c = \frac{1}{9}$, $E[X] = 2.25$. Working space

Q30 Medium Binomial 6m

EN: A fair coin is tossed 5 times. What is the probability of getting exactly 3 heads?

DE: Eine faire Münze wird 5 Mal geworfen. Wahrscheinlichkeit für genau 3 Mal Kopf?

Solution: $\frac{5}{16} = 0.3125$. Working space

Q31 Easy CLT 4m

EN: According to the Central Limit Theorem, the distribution of the sample mean approaches which distribution for large n ?DE: Welcher Verteilung nähert sich der Stichprobenmittelwert für großes n ?

A) Binomial B) Poisson C) Normal D) Exponential

Solution: C) Normal. Working space

Q32 Medium Normal 6m

EN: Given $\mu = 50$, $\sigma = 5$, find $P(X > 60)$.DE: Mit $\mu = 50$, $\sigma = 5$: Berechnen Sie $P(X > 60)$.**Solution:** 0.0228 (2.28%). Working space

E. Mechanics

Q33 Medium Projectile 7m

EN: A projectile is launched at 20 m/s at 30° . Find the time of flight ($g = 10$).DE: Ein Projektil mit 20 m/s unter 30° . Bestimmen Sie die Flugzeit ($g = 10$).**Solution:** 2 s. Working space

Q34 Medium Energy 6m

EN: A 2 kg object is dropped from 10 m. Find its impact velocity.

DE: Ein 2 kg Objekt fällt aus 10 m. Bestimmen Sie die Aufprallgeschwindigkeit.

Solution: $10\sqrt{2}$ m/s ≈ 14.14 m/s. Working space

Q35 Hard Rotation 9m

EN: A solid cylinder pulley $M = 4$ kg, $R = 0.5$ m has a 2 kg mass hanging from its string. Find the downward acceleration.DE: Vollzylinder-Rolle $M = 4$ kg, $R = 0.5$ m, hängende Masse 2 kg. Bestimmen Sie die Beschleunigung.**Solution:** 5 m/s². Working space

Q36 Medium SHM 7m

EN: A 1 kg mass is attached to a spring with $k = 100$ N/m. Find the period T .DE: 1 kg an einer Feder $k = 100$ N/m. Bestimmen Sie die Periode T .**Solution:** $\frac{\pi}{5}$ s ≈ 0.628 s. Working space

Q37 Easy Collision 4m

EN: In a perfectly inelastic collision, what happens to the colliding objects?

DE: Was passiert bei einem vollkommen unelastischen Stoß?

A) They separate B) They stick together C) Energy is conserved D) Speed doubles

Solution: B) They stick together. Working space

Q38 Medium Torricelli 7m

EN: Water flows from a hole 5 m below the surface. Find the exit velocity.

DE: Wasser fließt aus einem Loch 5 m unter der Oberfläche. Ausflussgeschwindigkeit?

Solution: 10 m/s. Working space

Q39 Hard Adiabatic 8m

EN: A monatomic ideal gas doubles its volume adiabatically. Find the pressure ratio ($\gamma = 5/3$).DE: Ein einatomiges ideales Gas verdoppelt sein Volumen adiabatisch. Druckverhältnis ($\gamma = 5/3$)?**Solution:** $2^{-5/3} \approx 0.315$. Working space

Q40 Medium Orbit 7m

EN: Derive the orbital velocity of a satellite at radius R in terms of GM .DE: Leiten Sie die Bahngeschwindigkeit eines Satelliten bei Radius R in GM her.**Solution:** $v = \sqrt{GM/R}$. Working space

Answer Key (Q1 – Q40)

Q	Answer	Q	Answer	Q	Answer	Q	Answer
Q1	$(x^2 - 3x + 2)^2$	Q11	$\frac{32\pi}{3}$	Q21	1, 3	Q31	C) Normal
Q2	$\frac{1}{x+1}$	Q12	1	Q22	$(7, 3, -4)$	Q32	0.0228
Q3	$(-\infty, -2)$	Q13	$1 - \frac{x^2}{2} + \frac{x^4}{24}$	Q23	C) $x + y + z = 1$	Q33	2 s
Q4	$\ln(x + \sqrt{x^2 + 1})$	Q14	$y = Ke^{x^2}$	Q24	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	Q34	$10\sqrt{2}$ m/s
Q5	$x = 1$	Q15	4.5, -6	Q25	$\frac{\sqrt{3}}{3}$	Q35	5 m/s ²
Q6	$\frac{23}{3}$	Q16	≈ 0.1667	Q26	2	Q36	$\frac{\pi}{5}$ s
Q7	$\frac{1}{6}$	Q17	$\frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}$	Q27	B) $\frac{10!}{2!2!2!}$	Q37	B) Stick together
Q8	$\frac{4}{\sqrt{3}} \times \frac{8}{3}$	Q18	2	Q28	0.4	Q38	10 m/s
Q9	$\frac{2}{9}$	Q19	$(4, -5, -6)$	Q29	$\frac{1}{9}, 2.25$	Q39	$2^{-5/3}$
Q10	-0.375 m/s	Q20	9	Q30	$\frac{5}{16}$	Q40	$\sqrt{GM/R}^{10}$

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- This study material is **not** an official ETH Zürich document. It is an independent preparation guide.
- **Questions:** All questions in this document are **original and adapted** problems in the ETH/Matura style.
- **References:** ETH Aufnahmeprüfung archive, GitHub *janikschuettler/eth-entrance-exam*, AKAD, Mathe-Physik-Aufgaben.
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